Vaishnavi Devi R

GitHub LinkedIn Portfolio Email 7200666243

PROFILE SUMMARY

AI and Data Science undergraduate with hands-on experience building machine learning models and LLM-powered applications, including RAG-based document retrieval and intelligent agentic AI systems. Skilled in Python, SQL, modern AI/ML frameworks, and backend development, with a strong focus on transforming data into practical, real-world solutions.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Frameworks/Libraries:** TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, LangChain, FastAPI, Pandas, NumPy
- **Tools:** Git, Docker, Jupyter Notebook, Postman, Power BI
- **Databases:** MySQL, PostgreSQL, Vector Databases (FAISS/Pinecone)
- **Concepts:** Machine Learning, Deep Learning, Natural Language Processing, Statistics & Probability, EDA, Feature Engineering, Predictive Modeling, Model Evaluation, Generative AI, Prompt Engineering, A/B Testing

PROJECTS

Intelligent Document Query & Retrieval System using LLM – GitHub

- Built an LLM-powered RAG system processing 500+ pages of documents with natural language query support.
- Structured documents into 1,000+ searchable segments through custom preprocessing and chunking.
- Integrated GPT-4 with a FastAPI backend for real-time responses in under 3 seconds.
- Improved contextual search relevance by ~40% over keyword-based search using a vector database.

Agentic AI System for Personal Productivity — Multi-Agent Voice Assistant with Task Automation – GitHub

- Designed and developed a 22-agent AI voice assistant where each agent performs specialized tasks including PC control, news fetching, stock tracking, web scraping, and email automation.
- Integrated Groq LLaMA 3.3-70B with a master orchestrator that understands natural language commands and routes them to the appropriate agent.
- Built a real-time voice pipeline with double-clap detection, wake-word recognition, and Windows SAPI voice replies for hands-free interaction.
- Developed a Flask dashboard displaying live agent statuses, activity logs, and a real-time command interface with secure API key management.

PROFESSIONAL EXPERIENCE

TechPrapt Private Limited — AI/ML Engineer Intern | *July 2026 – Present*

- Contributing to the development of an **AI-powered Recommendation System** for the AIEsphere platform.
- Working on **model evaluation, recommendation analysis, and performance optimization**.
- Conducting **UAT testing and validation** of AI/ML features to identify bugs and edge cases.
- Performing **benchmarking and comparative analysis** to evaluate and improve system performance.

CERTIFICATIONS

- Fundamentals of Generative AI – AWS — **Link**
- Generative AI Fundamentals – Databricks — **Link**
- Develop a Generative AI Chat App with Microsoft Foundry – Microsoft — **Link**

EDUCATION

B.Tech – Artificial Intelligence and Data Science (Integrated with Honours)
GKM College of Engineering and Technology, Anna University, Chennai
2023 – Present | **CGPA: 8.89**